

Derivatives Quiz

Calculus Version A

Name: _____

1. (4 pts) Let $f(x) = x^4 - 8x^2 + 7$. Find $f'(x)$.

2. (10 pts) Let $f(x) = x^4 - 8x^2 + 7$. Answer each question below.

a. Find $f'(x)$.

b. Find $f''(x)$.

c. Find all critical points of f .

d. Classify each critical point as a local min, local max, or neither.

3. (6 pts) Given $x^2 + y^2 = 25$, use implicit differentiation to find $\frac{dy}{dx}$.

4. (8 pts) Given $2x^2 + xy + y^3 = 10$, use implicit differentiation to find $\frac{dy}{dx}$.